ASSIGNMENT 5

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

dataset = pd.read_csv("C:/Users/admin/Desktop/CO3099/Social_Network_Ads.csv")

df = dataset.copy()
df.head()

df.info()

df.isnull().sum()

if 'Gender' in df.columns:
    df['Gender'] = df['Gender'].map({'Male': 1, 'Female': 0})

df.head()

if 'User ID' in df.columns:
    df = df.drop(['User ID'], axis=1)

df.head()

df.cov()

X = df.iloc[:, :-1].values
Y = df.iloc[:, -1].values

from sklearn.model_selection import train_test_split
```

```
xtrain, xtest, ytrain, ytest = train_test_split(
    X, Y, test_size=0.25, random_state=0
)

from sklearn.preprocessing import StandardScaler

sc = StandardScaler()
xtrain = sc.fit_transform(xtrain)
xtest = sc.transform(xtest)
from sklearn.linear_model import LogisticRegression

logreg = LogisticRegression()
logreg.fit(xtrain, ytrain)
y_pred_test = logreg.predict(xtest)
y_pred_train = logreg.predict(xtrain)

from sklearn.metrics import (
    confusion_matrix,
    accuracy_score,
    precision_score,
    recall_score
)

# Confusion Matrix
cm = confusion_matrix(ytest, y_pred_test)
print("Confusion Matrix:\n", cm)

# Accuracy
print("Test Accuracy:", accuracy_score(ytest, y_pred_test))
print("Train Accuracy:", accuracy_score(ytrain, y_pred_train))
```

```
# Precision
print("Precision:", precision_score(ytest, y_pred_test))


# Recall
print("Recall:", recall_score(ytest, y_pred_test))
```